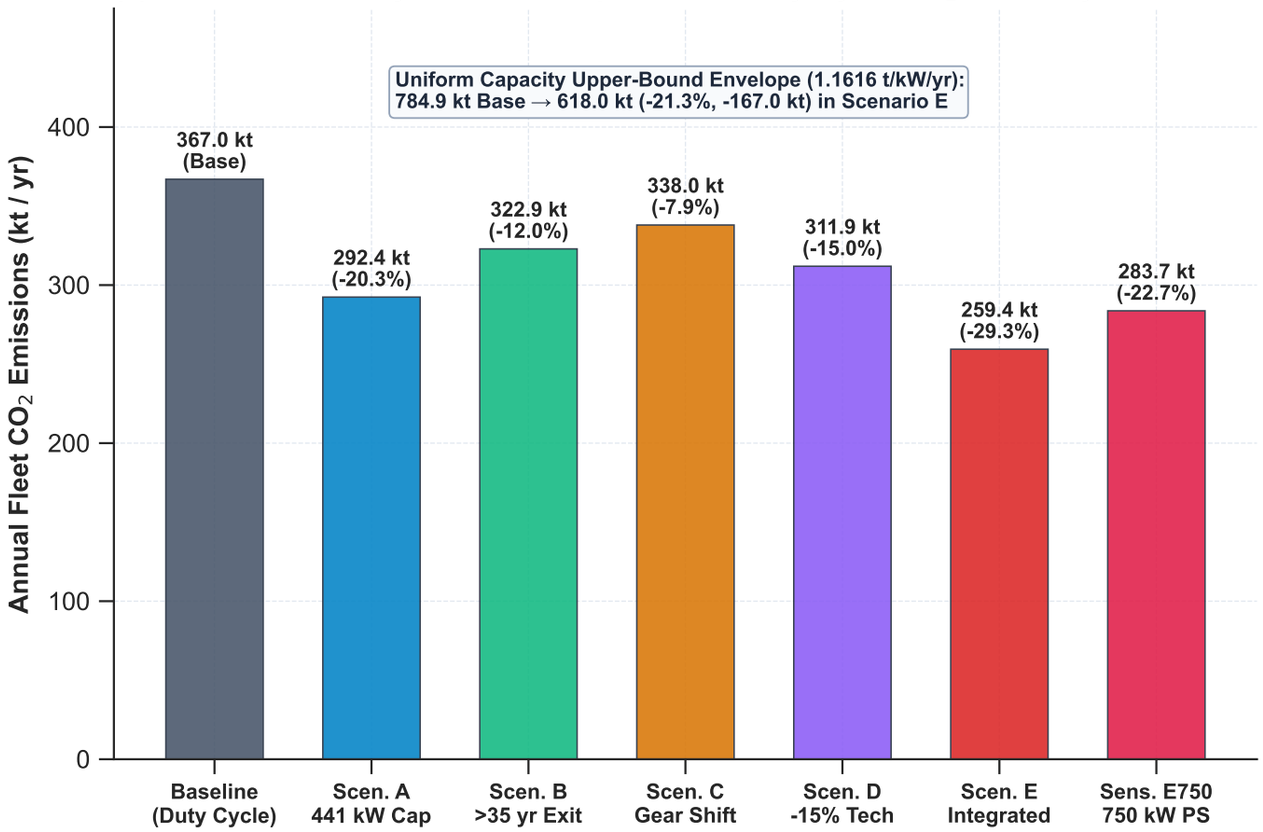
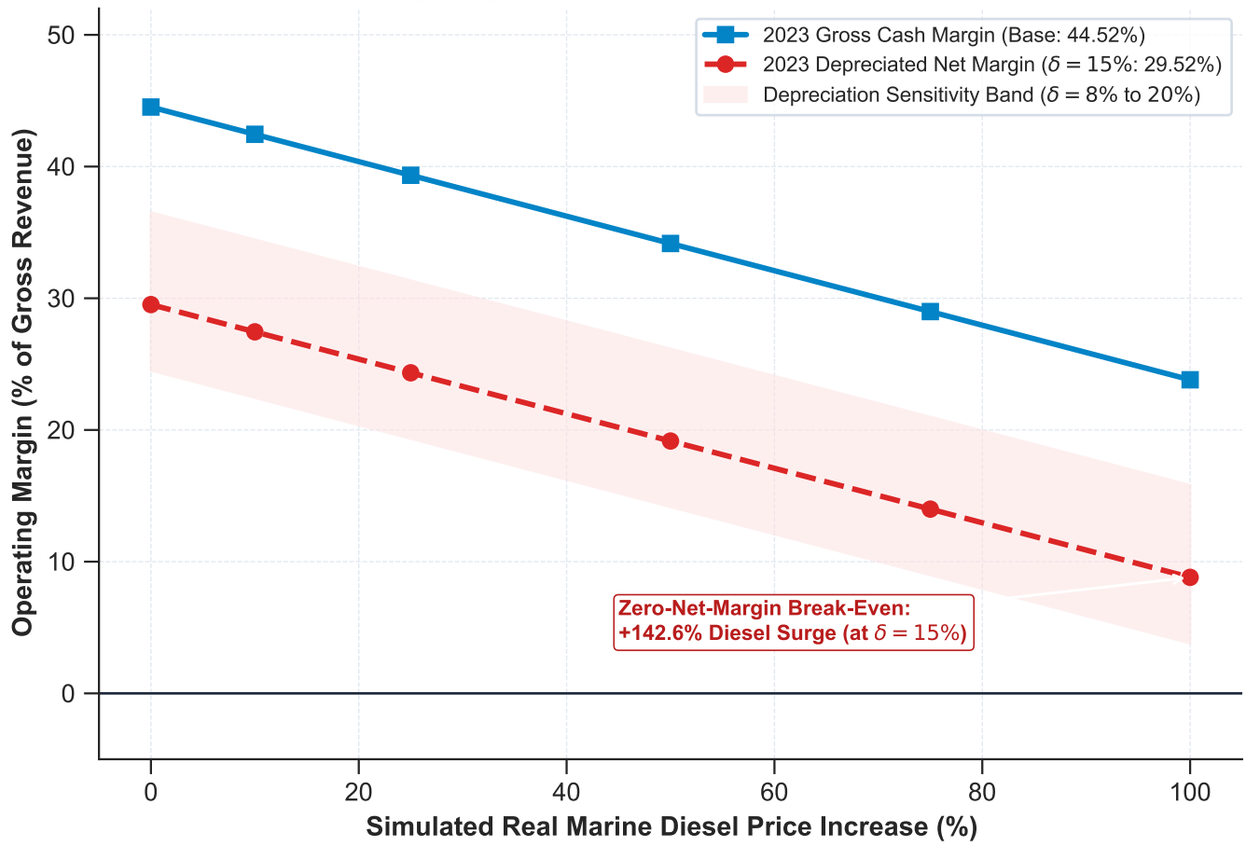
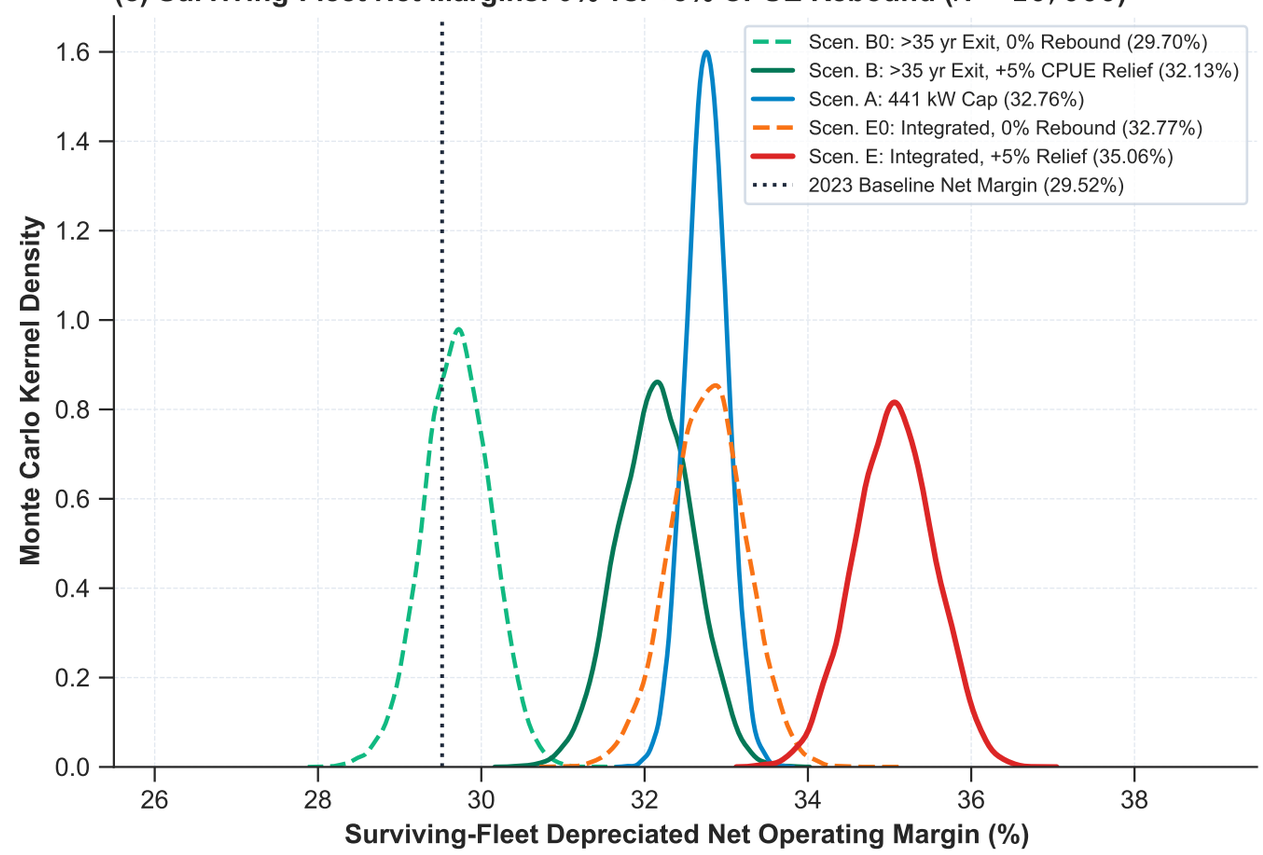
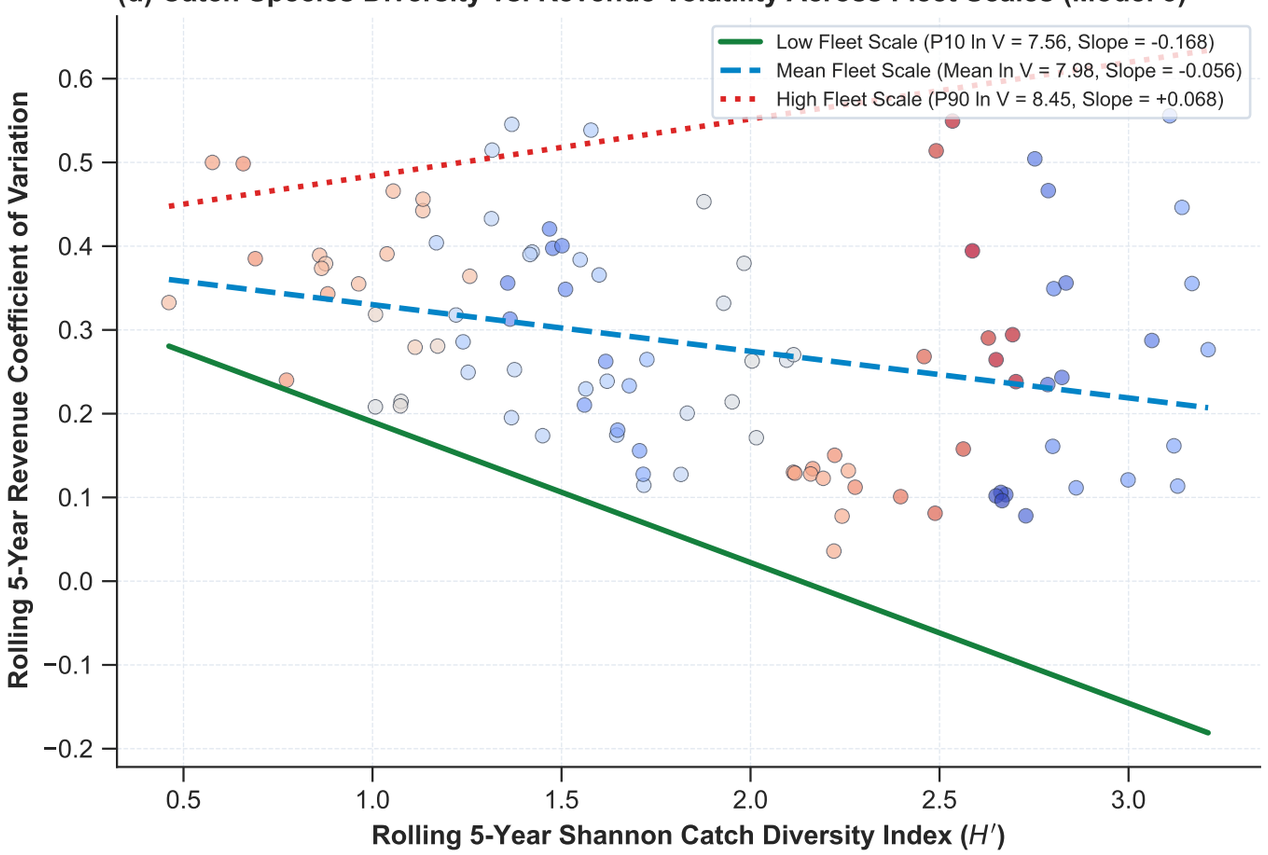

(a) Conditional Policy Scenarios: Operational Duty-Cycle $CO_2$ Inventory
Uniform Capacity Upper-Bound Envelope (1.1616 t/kW/yr):
784.9 kt Base → 618.0 kt (-21.3%, -167.0 kt) in Scenario E
Annual Fleet $CO_2$ Emissions (kt / yr)
0
100
200
300
400
367.0 kt (Base)
292.4 kt (-20.3%)
322.9 kt (-12.0%)
338.0 kt (-7.9%)
311.9 kt (-15.0%)
259.4 kt (-29.3%)
283.7 kt (-22.7%)
Baseline (Duty Cycle)
Scen. A 441 kW Cap
Scen. B >35 yr Exit
Scen. C Gear Shift
Scen. D -15% Tech
Scen. E Integrated
Sens. E750 750 kW PS


(b) 2023 TÜİK Operating Margins Under Diesel Price Shocks
2023 Gross Cash Margin (Base: 44.52%)
2023 Depreciated Net Margin ($\delta$ = 15%: 29.52%)
Depreciation Sensitivity Band ($\delta$ = 8% to 20%)
Zero-Net-Margin Break-Even:
+142.6% Diesel Surge (at $\delta$ = 15%)
Operating Margin (% of Gross Revenue)
0
10
20
30
40
50
0
20
40
60
80
100
Simulated Real Marine Diesel Price Increase (%)


(c) Surviving-Fleet Net Margins: 0% vs. +5% CPUE Rebound ($N = 10,000$)
Scen. B0: >35 yr Exit, 0% Rebound (29.70%)
Scen. B: >35 yr Exit, +5% CPUE Relief (32.13%)
Scen. A: 441 kW Cap (32.76%)
Scen. E0: Integrated, 0% Rebound (32.77%)
Scen. E: Integrated, +5% Relief (35.06%)
2023 Baseline Net Margin (29.52%)
Monte Carlo Kernel Density
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
26
28
30
32
34
36
38
Surviving-Fleet Depreciated Net Operating Margin (%)


(d) Catch Species Diversity vs. Revenue Volatility Across Fleet Scales (Model 3)
Low Fleet Scale (P10 ln V = 7.56, Slope = -0.168)
Mean Fleet Scale (Mean ln V = 7.98, Slope = -0.056)
High Fleet Scale (P90 ln V = 8.45, Slope = +0.068)
Rolling 5-Year Revenue Coefficient of Variation
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.5
1.0
1.5
2.0
2.5
3.0
Rolling 5-Year Shannon Catch Diversity Index ($H'$)